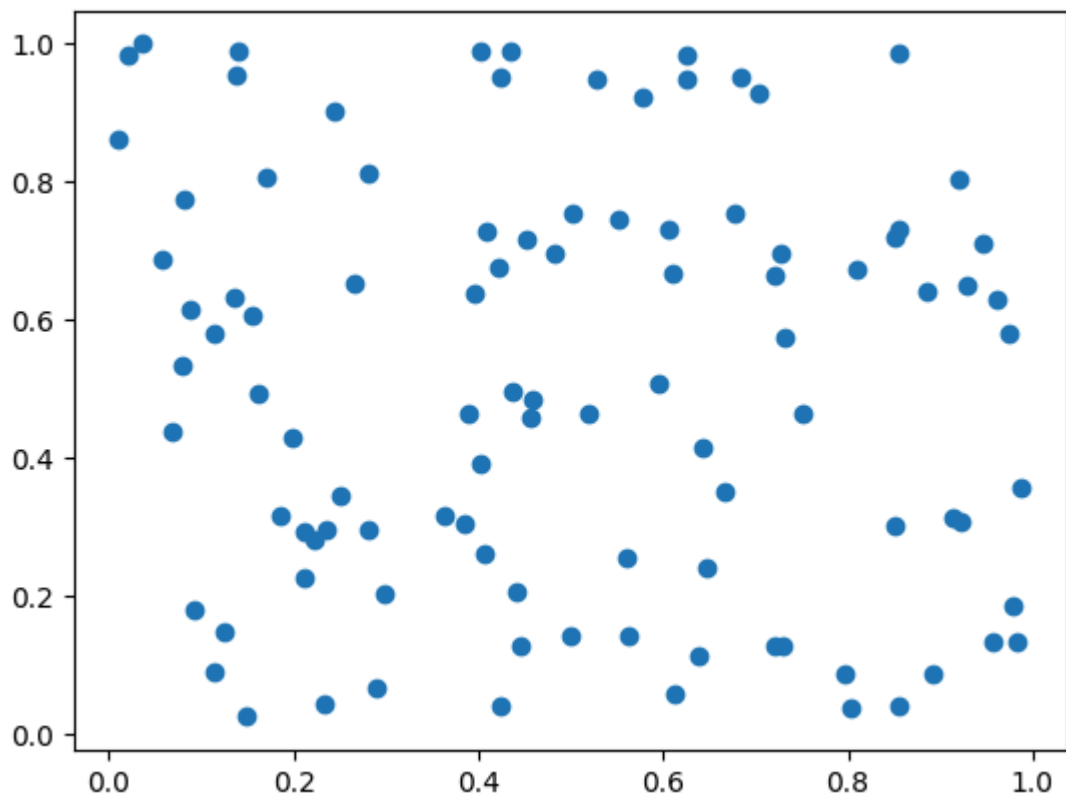


```
In [5]: import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score # check similarities - evaluation m
```

```
In [16]: x = np.random.rand(100,2)
```

```
In [17]: plt.scatter(x[:,0],X[:,1])
plt.show()
```



```
In [28]: kmeans = KMeans(n_clusters=3,random_state=0)
kmeans.fit(x)
```

Out[28]:

▼ KMeans ⓘ ?

► Parameters

► Fitted attributes

► 3 features 📄

```
In [29]: centroids = kmeans.cluster_centers_
labels = kmeans.labels_ # Model assigning the labels
```

```
In [30]: print(centroids)
print(labels)
```

```

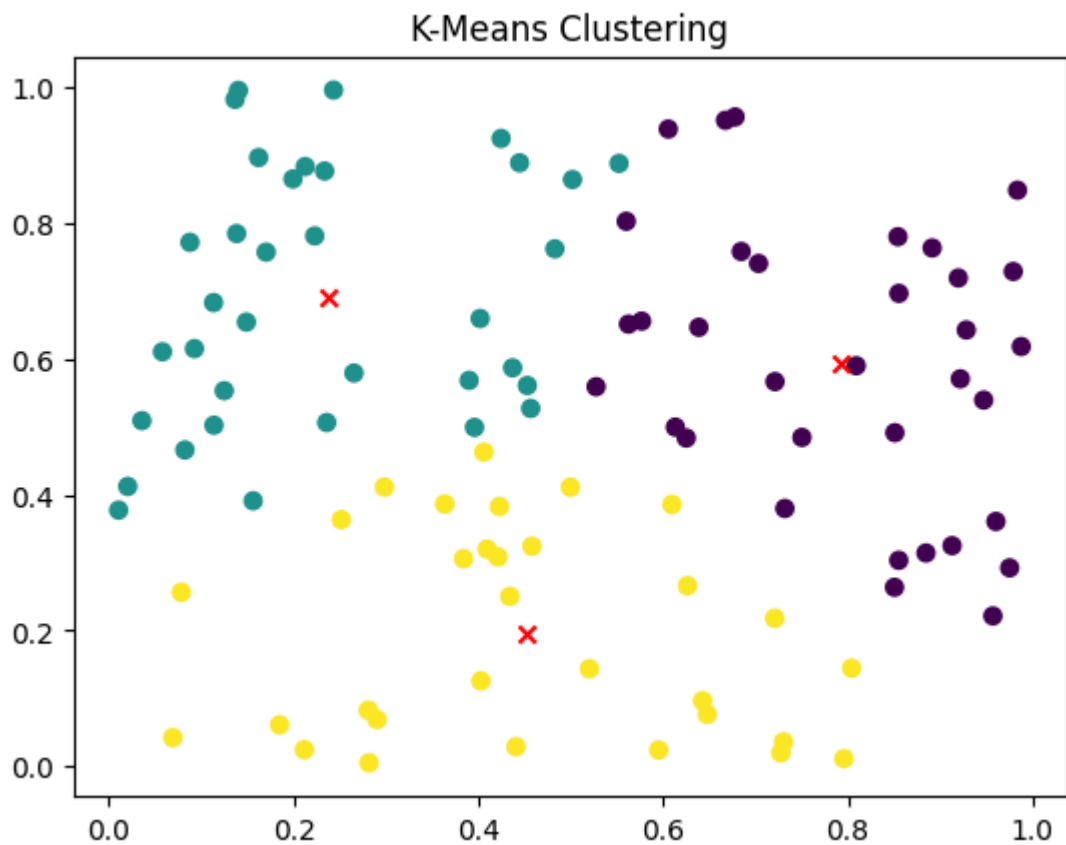
[[0.79322596 0.59214158]
 [0.23851183 0.69042655]
 [0.45196368 0.19496891]]
[0 1 1 1 2 0 2 1 0 1 2 1 1 0 2 0 0 1 0 1 0 0 2 0 1 0 1 2 2 1 0 0 1 1 0 1 0
 0 0 2 1 1 1 0 0 2 1 0 2 2 1 0 2 0 0 2 0 1 1 0 1 2 2 1 0 2 2 1 2 2 2 1 1 0
 0 1 0 2 2 2 1 0 0 1 0 1 1 2 2 2 2 1 2 2 1 2 2 0 0 1]

```

```
In [31]: print(f"Silhouette Score: {silhouette_score(x , labels):.4f}")
```

Silhouette Score: 0.3729

```
In [32]: plt.scatter(x[:,0],x[:,1],c=labels)
plt.scatter(centroids[:,0], centroids[:,1] , marker='x' , color='red')
plt.title("K-Means Clustering")
plt.show()
```



```
In [ ]:
```